

Eco-evolutionary context modifies a destructive plant invader's response to climate

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Summary

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- Understanding the relationship between climate and fitness will be important when predicting how plant populations respond to climate change. We conducted a replicated common garden experiment (4 sites × 2 yr) with 96 genotypes ($n = 22\,492$ individuals) of the invasive annual grass *Bromus tectorum* (cheatgrass) to understand how eco-evolutionary context mediates this important relationship.
- We grew genotypes in varying soil microclimate conditions by manipulating surface albedo, and at two planting densities. We calculated a 'climate mismatch' for each genotype within each microclimate to characterize the role of local adaptation to climate in explaining variation in fitness.
- Genotype-by-environment interactions increased the predictive accuracy of our statistical model of cheatgrass fitness with strong evidence for local adaptation to source climate. Survival responses to soil microclimate were density-dependent and there was stronger evidence for local adaptation at lower compared with higher planting density. Our model generally predicts increases in fitness with an increase in temperature across source populations.
- Important eco-evolutionary context (e.g. genotype-by-environment interactions, density dependence) should be considered when predicting plant fitness in an era of rapid environmental change.

Introduction

An ongoing challenge in ecology is predicting plant fitness in response to climate change, despite several decades of research devoted to this topic (Anderson, 2016; Anderegg, 2023; Perret *et al.*, 2024). Ecologists have historically tried to separate the

abiotic (e.g. climate) and biotic (e.g. population density) drivers of population dynamics and fitness, for example, in frameworks like the fundamental vs realized niche (Hutchinson, 1957), environmental filtering vs competitive exclusion (Kraft *et al.*, 2015), and direct vs indirect effects of climate change (Chu *et al.*, 2016). However, if we continue to assume that the biotic and abiotic

factors that influence population dynamics can be so neatly separated, we risk missing important interactions between these factors. A more integrated approach accounting for both ecological and evolutionary context may be needed to predict population dynamics in the face of climate change (Fig. 1).

Accounting for evolutionary context may be one way to improve predictions (Fig. 1a; Anderson & Song, 2020; Bontrager *et al.*, 2020). Plant genotypes can respond differently to the same environmental driver (Bradshaw, 1965). These genotype-by-environment (i.e. 'G × E') interactions are ubiquitous across plant taxa (Marais *et al.*, 2013) and can account for significant variation in plant functional traits. For example, coastal marsh sedges exhibited strong genotype-by-CO₂ (i.e. ambient or elevated) interactions in aboveground and belowground biomass in a large mesocosm experiment (Vahsen *et al.*, 2023). When ignored, genotype-by-environment interactions could decrease the accuracy of predictions of population performance, especially for species with high standing genetic variation. Local adaptation – a specific form of genotype-by-environment interaction such that the fitness of a genotype at its home site is higher than fitness of foreign genotypes (Kawecki & Ebert, 2004) – can emerge over time, which could increase the predictability of genotypic responses to climate when source environmental conditions are known (DeMarche *et al.*, 2019). However, under climate change, this increased predictability may be less reliable because local adaptation could hamper populations if environmental conditions are especially novel. This could lead to fitness declines unless adaptive evolution can keep pace with shifts in the abiotic environment (Etterson & Shaw, 2001).

Additionally, intraspecific density dependence has long been recognized as a key biotic factor driving population dynamics (Andrewartha & Birch, 1954; Nicholson, 1954); however, how it mediates climate responses is less well understood (Postma *et al.*, 2021). Climate change may shift the strength and direction of density-dependent regulation (Fig. 1b). For example, the negative effects of competition could be exacerbated under climate change (Jaatinen *et al.*, 2021) or, alternatively, be ameliorated if

the effects of climate change become more significant than those of negative density dependence (Ouyang *et al.*, 2014). Climate change could additionally promote facilitation if environmental conditions are particularly stressful (Schultz *et al.*, 2023).

Evolutionary and ecological contexts might interact in complicated ways. Theory suggests that locally adapted genotypes might outperform others only when densities are high, or local advantages might be reduced at higher densities if the effect of local adaptation is 'swamped' by stronger effects of intraspecific competition (Fig. 1c; Gomulkiewicz *et al.*, 1999; Filin *et al.*, 2008; Burgess *et al.*, 2013). Empirical tests of this theory are rare, but Siepielski *et al.* (2016) found that density-dependent competition and local adaptation were coupled in experimental damselfly populations. However, the causality in their experiment was unclear, because the strength of local adaptation could mediate the strength of density-dependent competition, or vice versa. Notably, the negative effects due to local adaptation in this damselfly experiment often outweighed those of density dependence. More generally, the relative importance of density dependence, local adaptation, and their interaction remains unknown.

Invasive plants are an apt system for investigating the eco-evolutionary context of the relationship between climate and fitness. Species invasions are often referred to as natural evolutionary experiments (Moran & Alexander, 2014); invasive populations encounter novel environments as they establish and spread throughout their introduced range. Invasive plants' ability to outcompete resident native species may reflect, in part, their ability to plastically respond to shifts in their environment (Davidson *et al.*, 2011). Population densities also shift rapidly during invasion, with densities typically lowest at the invasion's leading edge (Grayson & Johnson, 2018). Despite the relatively short introduction histories of invasive plants in North America (i.e. on the orders of decades to a century), there is substantial evidence for local adaptation (Colautti & Barrett, 2013; Oduor *et al.*, 2016; Gamba *et al.*, 2025), suggesting that invasive plants may develop strong genotype-by-environment interactions within and/or among populations. We use the term 'local

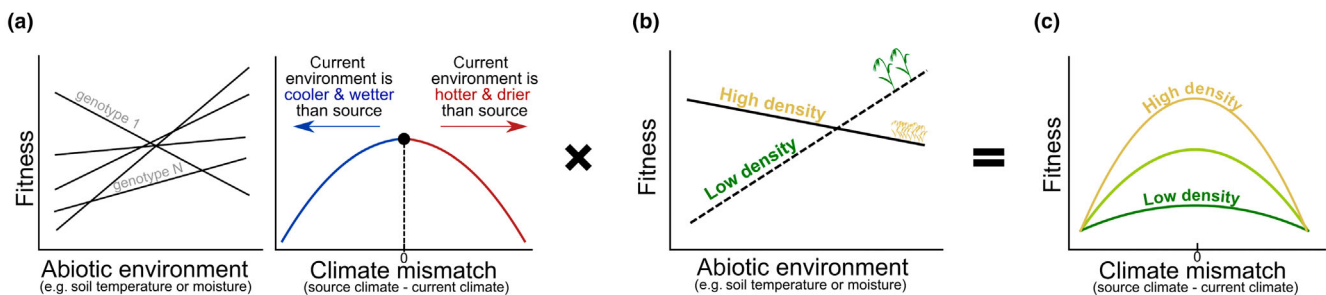


Fig. 1 Abiotic environment–fitness relationships can be shaped by evolutionary (a), ecological (b), and eco-evolutionary (c) contexts. (a) Genotype-by-environment interactions can obscure climate–fitness patterns when genotypes vary in their environmental responses. Fitness across a gradient of climate mismatch, measured as the distance between a genotype's source and current environment, can suggest local adaptation if fitness declines from its optimum at climate mismatch = 0 (denoted with a black point) with increasing mismatch in both the positive and negative direction. (b) Ecological context, such as intraspecific density-dependent responses to climate, can further influence fitness outcomes. Here, we show stronger responses for plants at low density than plants at high density, but we note that the opposite prediction is equally likely. (c) Interactions between these processes may occur, for example, when the strength of local adaptation depends on intraspecific population density.

adaptation' in its operational evolutionary sense, recognizing that for invasive species, adaptation to novel environments can occur over rapid timescales. In sum, understanding the interactions between climate, density, and local adaptation across an expanding range could help predict the spread of invasive species.

However, quantifying the roles of ecological, evolutionary, and eco-evolutionary context in determining fitness under climate change is challenging. For example, to understand how an invasive plant species, or more generally any species, responds to shifts in its environment over space and time, phenotypic variation should be characterized for a large and representative sample of genotypes (Saltz *et al.*, 2018). Testing for local adaptation is particularly difficult because, ideally, it requires experiments that compare several genotypes' fitness responses across many different environments (Kawecki & Ebert, 2004; Blanquart *et al.*, 2013). However, methodology developed for forestry provenance trials can also be used to test for evidence of local adaptation, specifically by calculating the deviation between the local weather experienced by a plant in a trial and the climate norm of the plant's genotypic origin. If populations are locally adapted to climate in their introduced range, larger positive or negative values of this 'mismatch' should result in reduced fitness (Fig. 1a; Wang *et al.*, 2010; Capblancq *et al.*, 2023; Villoutreix *et al.*, 2025).

We report the results of a large, 2-yr, replicated common garden experiment designed to address these challenges. While motivated by general questions about eco-evolutionary context, our case study focused on the destructive invasive annual grass, *Bromus tectorum*, commonly known as cheatgrass. Since its introduction in the late 1800s, *B. tectorum* has reached substantial ($\geq 15\%$) cover over nearly a third of the Intermountain West (Bradley *et al.*, 2018). *Bromus tectorum* plants outcompete native vegetation, in part, by accelerating fire cycles, because they can successfully establish in highly disturbed environments (Pelant, 1996). Cheatgrass can reach very high densities in the Intermountain West (up to 15 000 plants m^{-2} ; Stewart & Hull, 1949), resulting in near-monoculture stands in previously native perennial-dominated habitats. Introduced populations have significant genetic variation (Leger *et al.*, 2009), even with a predominantly selfing and cleistogamous (i.e. closed-flower) mating system (Novak & Mack, 2016; Gamba *et al.*, 2025). A recent genomic study provides strong evidence for local adaptation to climate in the introduced range (Gamba *et al.*, 2025). In our study, we focused on two primary axes of climate variation – soil temperature and moisture – because these are both projected to change substantially by the end of the century (Joo *et al.*, 2020; Soong *et al.*, 2020) and also be important drivers of *B. tectorum* population dynamics (Terry *et al.*, 2024).

We addressed two main research questions: First, do *B. tectorum* genotypes from across the introduced range vary in their demographic responses to climate, as evidenced by genotype-by-environment interactions? We have previously shown that intraspecific density, temperature, and genotype independently, but not interactively, influenced flowering phenology (Vahsen *et al.*, 2025). However, because fitness integrates the effects of multiple traits, we expected a greater likelihood of

strong genotype-by-environment interactions. Given evidence for such variation, we then ask if it is related to genotypes' climate of origin within the introduced range, suggesting widespread local adaptation to climate (Fig. 1a). Second, are ecological or evolutionary responses to climate mediated by intraspecific density dependence (Fig. 1b,c)? We predicted these aspects of eco-evolutionary context would explain variation in the relationship between soil microclimate and *B. tectorum* fitness.

Materials and Methods

Common garden experiment

We conducted a common garden experiment with replicates of 96 *Bromus tectorum* L. genotypes that were collected in the field in 2020, with collections aimed to capture climatic variation across Western North America (Supporting Information Fig. S1; Table S1). Genotypes were propagated in a glasshouse for one generation following collection to standardize potential maternal effects (see Vahsen *et al.*, 2025 for details). We refer to all seeds that originated from the same field-collected plant as genotype replicates.

The common garden experiment was replicated across four sites in the Western United States (Fig. 2a; Sheep Station, ID [44.246°N, −112.214°E], elevation = 1647 m; Baltzor, ID [43.208°N, −116.995°E], elevation = 1573 m; Wildcat, ID [43.474°N, −116.902°E], elevation = 823 m; Cheyenne, WY [41.212°N, −104.852°E], elevation = 1915 m) and two discrete growing seasons (fall 2021–summer 2022, fall 2022–summer 2023; hereafter 2022 and 2023 growing seasons, respectively). Unless explicitly noted, we followed the same methods at every site. In each growing season and common garden, we manipulated planting density (two levels) and soil temperature (two levels) in a fully factorial design (Fig. 2c). We assigned density and soil temperature treatments at the plot level using a randomized block design, such that each block ($n = 10$) had one plot of each soil temperature by density combination (10 blocks $\times$ 2 temperature treatments $\times$ 2 density treatments = 40 plots per site $\times$ year combination).

In fall 2021, we planted 100 *B. tectorum* seeds in each plot. Replicates of genotypes were distributed across plots in a stratified random design that maximized the number of plots each genotype was represented in within the constraints of the available replicates of each genotype from the glasshouse propagation. Individual seeds were planted 1 cm below the soil surface, glued to a toothpick with the lemma oriented upward to permit normal *B. tectorum* development while allowing us to track individual plants over the course of the growing season, as they remain close to the toothpick throughout their life cycle (Leger *et al.*, 2009). We manipulated density by changing the size of the plots (Fig. 2c). The high-density plots (2500 plants m^{-2}) were 1.2 m $\times$ 1.2 m and the low-density plots (100 plants m^{-2}) were 1.5 m $\times$ 3 m. Our low-density treatment represents a density below the lower range of those observed in a historical survey in the invasive range (1076–15 069 plants m^{-2} ; Stewart & Hull, 1949) and is likely insufficient to outcompete native

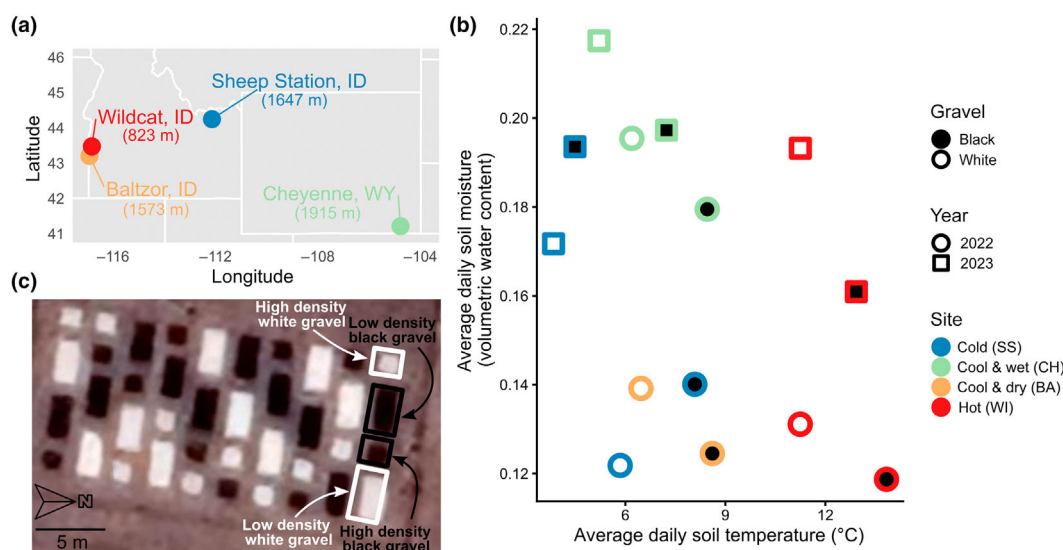


Fig. 2 Replicated common garden experimental design for *Bromus tectorum* genotypes across two discrete growing seasons. (a) Map showing the four common garden locations colored based on their distinguishing soil microclimate characteristics in panel (b): cold (SS) = Sheep Station, ID [44.246°N, −112.214°E], cool & wet (CH) = Cheyenne, WY [41.212°N, −104.853°E], cool & dry (BA) = Baltzor, ID [43.208°N, −116.995°E], and hot (WI) = Wildcat, ID [43.474°N, −116.902°E]. (b) Average daily soil temperature and volumetric water content recorded continuously throughout the growing seasons using soil probes. Data are shown for two growing seasons (2022 = 2021–2022; 2023 = 2022–2023), with points colored by gravel treatment (white = high albedo = lower temperature; black = low albedo = higher temperature) and outlined by site. Environmental data for the Baltzor site are only shown for the 2022 season. (c) Aerial photograph of the Cheyenne, WY common garden captured on 12 July 2022 (Google Earth, 2026) showing square, high-density and rectangular, low-density plots assigned to white or black gravel treatments (higher vs. lower albedo, respectively).

perennial species (Mazzola, 2008). Our high-density treatment represents a density well within densities observed in the invasive range and was primarily constrained by the minimum spacing needed to track and separate individual plants.

Planting areas within the plots were smaller to account for potential edge effects: the high-density planting area was 0.2 m × 0.2 m, with plants spaced 2 cm apart, and the low-density planting area was 0.5 m × 2 m, with plants spaced 10 cm apart. Within the extended space beyond the planting area within a plot, we planted two rows of border *B. tectorum* seeds collected from plants adjacent to each common garden site to further reduce edge effects on experimental plants. These plants were spaced according to density treatment (low density = 10 cm, high density = 2 cm). Due to limited seed availability for the 2023 growing season, we seeded only 90 plants in each plot at Wildcat, ID and 80 plants in each plot at Sheep Station, ID and Cheyenne, WY, but the distances between the plants in each density treatment were kept the same as in the 2022 growing season. In 2023 at the Baltzor, ID site, frost-heaving significantly disturbed our plant markers, preventing us from tracking individual plants confidently; we only present data from that common garden for 2022.

We manipulated soil temperature by distributing black gravel (low albedo, high temperature) and white gravel (high albedo, low temperature) over bare soil across all of the plots (see Maxwell *et al.*, 2023; Vahsen *et al.*, 2025 for details). This gravel treatment resulted in a *c.* 2°C difference in soil surface temperature between the white and black gravel treatments across the growing seasons (Fig. 2b,c). Before adding gravel to each of the

plots, we created the bare soil beds by applying herbicide (glyphosate) as well as manual weeding, shallow tilling, and leveling. Given these methods, we did not eliminate potential existing seed banks of cheatgrass individuals, but our toothpick method, described above, allowed us to confidently track experimental individuals vs those that may have emerged from a seedbank.

We recorded emergence, flowering phenology (Vahsen *et al.*, 2025), and survival for each individual plant across each growing season. Throughout the growing season, we hand-weeded plots to remove resident *B. tectorum* and other species, while maintaining the *B. tectorum* border rows. At the end of the growing season, we harvested individual plants or, where necessary, individual inflorescences as their seedheads matured (i.e. once they were purple in color). Harvesting *B. tectorum* plants within a short maturation window proved challenging given our large sample size, which led to some plants dropping seeds before harvest (i.e. recorded as ‘seed drop’) and some plants being harvested at a different maturation status compared with others (i.e. more green florets than purple). We accounted for this variability, in part, by removing observations with recorded ‘seed drop’ from our analyses and by pooling multiple harvests of a plant into a single value for seed count. Given our large sample size ($n = 22\,492$ individual plants), observational errors in seed count due to challenges in harvesting likely do not affect our inference.

To measure individual plant fecundity across all common gardens in the 2022 growing season, we separated vegetative and inflorescence mass for each plant. Many plants in the experiment produced hundreds to thousands of seeds, making counts of individual seeds infeasible. For plants with > 50 total seeds, we

subsampled 50 filled seeds and their spikelets from the inflorescence mass. Based on the total inflorescence mass, the subsampled inflorescence mass, and the known number of subsampled seeds, we estimated the total number of seeds on the plant (total number of seeds = [50 seeds × inflorescence mass]/subsampled inflorescence mass). Given that the strong relationship between inflorescence mass (g) and the number of seeds derived from the 2022 data was invariant across gravel and density treatments (Fig. S2), we did not count seeds in 2023. Instead, we estimated the number of seeds from total inflorescence mass using a regression fit to the 2022 data, for which we had both counts of the number of seeds and measurements of inflorescence mass for each plant:

$$\log_e(\text{number of seeds}) = 5.927 + 0.923 \times \log_e(\text{inflor. mass})$$

for all plants with a non-zero inflorescence mass ($R^2 = 0.94$; Fig. S2).

Characterizing soil microclimate

To estimate average daily soil temperature and moisture (volumetric water content (vwc); Fig. 2b), we first calculated the average daily minimum and maximum values for each day across four EM50 data loggers with 5TM temperature probe sensors (Decagon, Pullman, WA, USA) inserted 5 cm vertically into the soil, which encompasses a significant proportion of *B. tectorum* seedling root volumes. Soil temperature and moisture measurements were integrated across the 0–5 cm soil depth. One probe was placed in each of four randomly selected plots for both black and white gravel treatments at each common garden. We then calculated the average daily values for both soil temperature and moisture as the mean of the average daily minimum and maximum values. We obtained average daily soil temperature and moisture for multiple temporal windows within each growing season to inform our hypotheses about how temperature affects demographic rates of interest: January–May for plant emergence and survival to reproduction, and March–May for fecundity. In the statistical analyses described below, we modeled survival and fecundity as a function of these two continuous soil microclimate variables rather than using sites, treatments, and years as categorical variables.

Calculating climate mismatch index

To assess the strength of evidence for local adaptation, we calculated an index of ‘climate mismatch’ for each plant as the difference between weather experienced by a genotype in the experiment (as informed by its common garden location and growing season; $n = 7$) and the 30-yr climate norm of the source location from which the genotype was collected ($n = 96$; Fig. 1a). This mirrors a common technique for detecting local adaptation in forestry provenance trials (Wang *et al.*, 2010; Capblancq *et al.*, 2023; Villoutreix *et al.*, 2025). We used Daymet data (Thornton *et al.*, 2022) to characterize the weather at the common garden sites during the experimental growing seasons and

the 30-yr climate norm (1991–2020) for each genotype source location.

The monthly Daymet weather and climate data were standardized (i.e. centered and scaled) and then reduced in dimension from 19 bioclimatic variables to one major principal component axis (hereafter PC 1; explains 36% of climatic variation), which represented a continuum from cooler and wetter sites to hotter and drier sites. For each unique genotype by common garden and growing season combination, we calculated the absolute difference between a genotype’s source PC 1 value and the PC 1 value of the weather that plant experienced during the experiment. Larger positive or negative climate mismatch values indicate that a replicate plant of a certain genotype is growing in conditions that are either hotter and drier or cooler and wetter, respectively, than the climate at the location where the genotype was originally collected (Fig. 1a).

Statistical analysis

Model structure We analyzed both survival and fecundity data for individual plants using a hierarchical Bayesian hurdle model (see Notes S1 for full model specification). We selected a hurdle model because it is a two-step model that allows for fixed and random effects to separately influence the different component vital rates of fitness. Here, we focus on two vital rates: the probability of surviving to reproduction (a binary response), and the number of seeds produced conditional on surviving to reproduction (i.e. seed count > 0). Thus, our full model included a Bernoulli likelihood to model a plant’s probability of survival to reproduction and a zero-truncated negative binomial likelihood to model a plant’s fecundity (number of seeds produced) conditional on surviving to reproduction. For the Bernoulli likelihood, plant survival to reproduction was a function of measured soil temperature, measured soil moisture, and intraspecific density treatment. We included linear and quadratic terms for ‘climate mismatch’ to test for evidence consistent with local adaptation (Fig. 1a). To test for density-dependent responses to climate (Fig. 1b,c), we included interactions between intraspecific density and soil temperature, soil moisture, and climate mismatch linear and quadratic terms. The random intercepts in the model were genotype, site × year × gravel combination (to account for exogenous variation that could not be explained by the soil temperature and moisture variables), and plot (nested within site and year). Finally, we included random slopes for soil temperature, soil moisture, and density treatment across genotypes to test for genotype-by-environment interactions. We used the same fixed and random effect model structure for a zero-truncated negative binomial likelihood fit to fecundity data (conditional on a plant surviving to reproduction), except that for characterizing intraspecific density, we calculated a continuous number of neighbors that surrounded each plant, which was informed by the number of surviving neighbors in the spring (mean = 41% and range 0–88% survival to reproduction across plots) within a fixed radius of a plant (see Methods S1).

We fit the Bayesian hierarchical model in Stan using a Hamiltonian Monte Carlo algorithm (Hoffman & Gelman, 2014) with

three chains, each with 1000 burn-in iterations and 2000 sample iterations, and based our inference on the output from all chains. We implemented our model via CMDSTANR (v.0.8.1; Gabry *et al.*, 2024) in R (v.4.4.2; R Core Team, 2024), and all additional analyses and figure generation were conducted in R. We used moderately informed priors for our analyses to stabilize model fitting and improve predictive performance (Hobbs & Hooten, 2015). To confirm convergence, we visually assessed trace and density plots of parameters and ensured all Gelman-Rubin statistics were < 1.01 (Gelman & Rubin, 1992). We assessed model fit by plotting the posterior predictive distributions for each observation against its observed value and conducting posterior predictive checks on the first two moments of the observed data distribution (Figs S3, S4).

Linking research questions with model inference To address our first question, concerning the importance of genotype-specific environmental responses, we fit two versions of the hierarchical model: one that included random slopes to capture the varying effect of treatments and covariates across genotypes (i.e. genotype-by-environment interactions) as well as the climate mismatch covariates, and one that did not include random slopes or climate mismatch covariates (a null model that assumes that all genotypes respond similarly to treatments and covariates). We used leave-one-out cross validation (Vehtari *et al.*, 2017) to select the models with the highest predictive accuracy for both the survival and fecundity submodels.

After finding evidence for strong genotype-by-environment interactions, we next asked whether a genotype's response to its current environment depended on its source environment. We calculated each genotype's sensitivity to soil temperature and moisture for survival, fecundity, and fitness *post hoc* by dividing the difference between the predicted values of each demographic rate at the minimum and maximum values of the environmental gradient by that environmental distance (i.e. a linear slope). We regressed each genotype's sensitivity to soil temperature and moisture against the source climate of the genotype as quantified by its value across PC 1 (continuum from cool and wet sites to hot and dry sites). A significant positive or negative slope from these regressions would indicate that a genotype's sensitivity to its current environment depends on its source environment. This approach allows us to assess the effect of source climate on sensitivity to soil temperature and moisture separately. To test for evidence consistent with local adaptation, we evaluated whether the 95% credible intervals for quadratic regression coefficients for climate mismatch in the Bernoulli (survival) and zero-truncated negative binomial (fecundity) models included 0.

Finally, to test for evidence of density-dependent responses to climate, we evaluated whether the 95% credible intervals for coefficients associated with interactions between density and soil temperature, density and soil moisture, as well as density and climate mismatch, included 0 in the full hierarchical model. While we gain direct inference from the regression parameters in the Bernoulli and zero-truncated negative binomial likelihoods on survival and fecundity, respectively, we also sought to understand how

genotype-by-environment interactions and density-dependent responses to climate affect fitness. To that end, we calculated the expected fitness of each plant as the model-derived quantity, $\text{fitness} = P(\text{survival to reproduction}) \times E(\text{fecundity})$, using the posterior samples of predicted survival and fecundity.

Predicting genotype-specific fitness responses to elevated temperatures We used our model to generate preliminary predictions of how the source populations of *B. tectorum* we studied might respond to climate change. To do this, we used our model to predict the mean natural-log fitness for each genotype source population under two conditions: present-day and an approximated future climate of $+2^{\circ}\text{C}$, which is a conservative projection for the expected change in soil temperature by the end of 2100 (Soong *et al.*, 2020). For this simulation exercise, we ignored changes in soil moisture and density and made predictions assuming average soil moisture and density across common garden sites. We assumed that present-day genotypes were locally adapted to their source climates (i.e. climate mismatch = 0). For predictions of future fitness, we modified the data in two ways: we added 2°C to the estimates of soil temperature and we predicted PC 1 values used in the climate mismatch metric calculation given a 2°C increase in both mean maximum and minimum temperature across all months (see Methods S2 for additional details). Finally, we compared our predictions from the full model as described above to one that did not account for genotype-by-temperature interactions or climate mismatch to investigate the role evolutionary context played in determining predicted fitness responses.

Results

Do genotypes vary in their response to climate?

We found strong evidence that accounting for genotype-by-environment interactions improved predictive accuracy in our models (Tables 1, S2). 'Strength of support' values reported in Table 1 were both > 2 , which is similar to a frequentist z -score suggesting there are differences in predictive accuracy that would be considered significant at the $\alpha = 0.05$ level (Vehtari *et al.*, 2017; Sivula *et al.*, 2023). Fecundity was more strongly influenced by genotype-by-environment interactions than survival (Table 1; strength of support for fecundity model = 6.8 vs survival model = 3.8). Genotype-specific survival exhibited similar variability in response to soil moisture (Bernoulli random slope SD $\sigma_{\text{G} \times \text{vwc}} = 0.13$, 95% credible interval [0.08, 0.18]) and temperature ($\sigma_{\text{G} \times \text{temp}} = 0.12$ [0.06, 0.17]), whereas genotype-specific fecundity exhibited greater variability in response to soil temperature (NB random slope SD $\omega_{\text{G} \times \text{temp}} = 0.12$ [0.09, 0.15]) compared with moisture ($\omega_{\text{G} \times \text{vwc}} = 0.07$ [0.04, 0.10]) in the common gardens.

Across the soil temperature gradient, genotype survival to reproduction exhibited a relatively consistent response (Fig. S5a), with average survival generally increasing as a function of temperature, although this overall trend was not significantly different from zero (Bernoulli regression coefficient $\beta_{\text{temp}} = 0.49$

Table 1 Comparison between the *Bromus tectorum* genotype-by-environment interaction hierarchical model and null hierarchical model (i.e. model that did not account for genotype-by-environment via random slopes for abiotic and biotic covariates across genotypes) performance.

Vital rate	Preferred model	Mean ELPD difference (SE)	Strength of support	$\sigma_{G \times \text{dens}}$ [95% CI]	$\sigma_{G \times \text{temp}}$ [95% CI]	$\sigma_{G \times \text{vwc}}$ [95% CI]
Survival	Random slope model > null model	−36.8 (9.6)	3.8	0.03 [0.00,0.08]	0.12 [0.06,0.17]	0.13 [0.08,0.18]
Fecundity	Random slope model > null model	−111.4 (16.3)	6.8	0.06 [0.02,0.09]	0.12 [0.09,0.15]	0.07 [0.04,0.10]

Comparisons are made for both the survival and fecundity portions of the model via leave-one-out cross validation (Vehtari *et al.*, 2017). 'Mean ELPD' represents the mean expected log pointwise predictive density, where higher values represent stronger support for a model. 'Strength of support' is calculated by dividing the mean ELPD difference by its SE, where values > 2 indicate meaningful differences in the models. Means and 95% credible intervals for each of the random slope SD are shown for the preferred random slope submodels. 'G', genotype; 'dens', density; 'temp', soil temperature; 'vwc', volumetric soil water content.

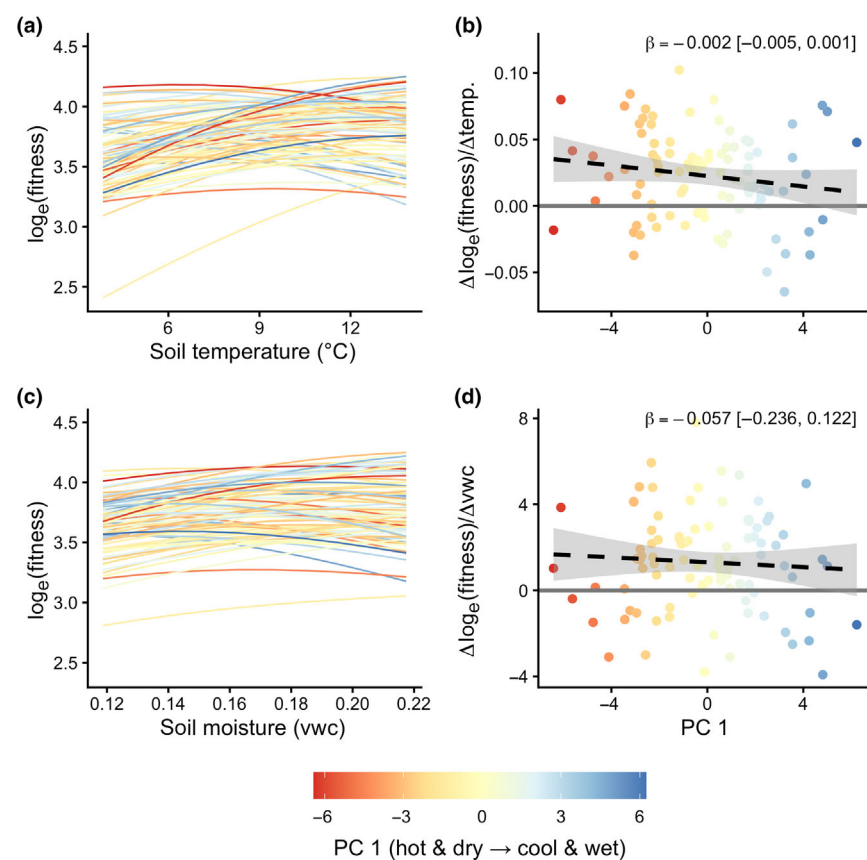
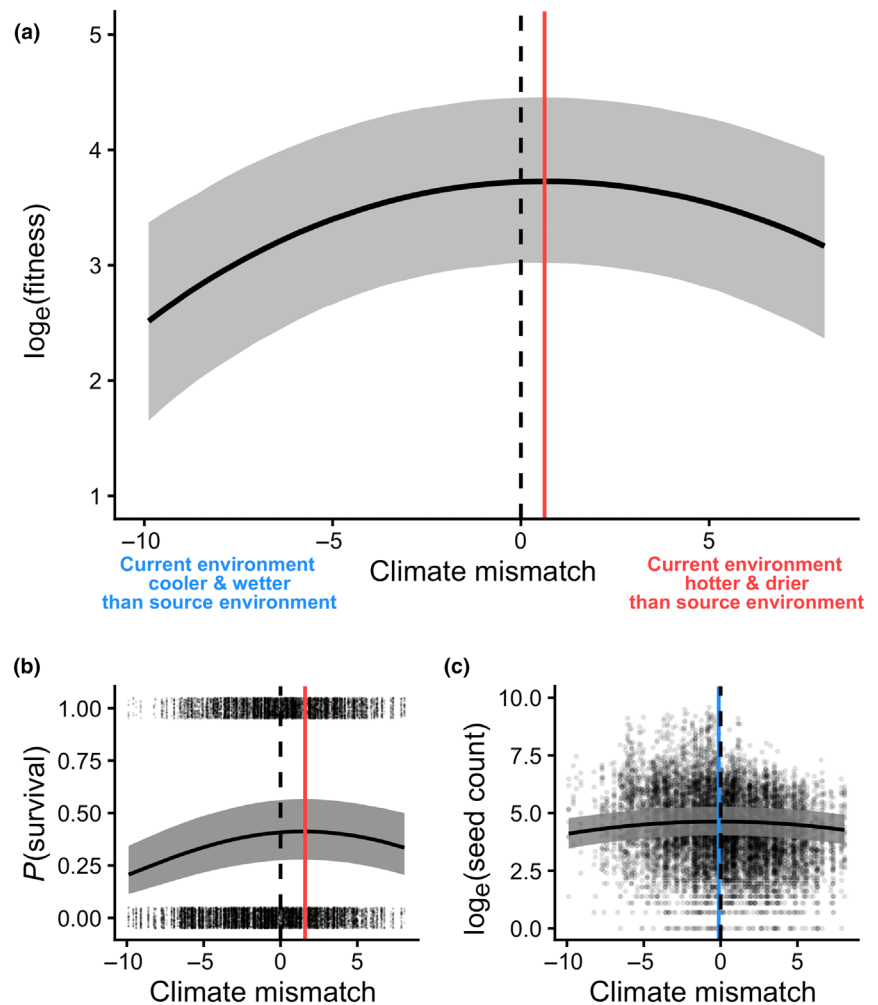


Fig. 3 Predicted fitness (calculated as the probability of survival to reproduction $\times$ the number of seeds produced) of *Bromus tectorum* genotypes grown in replicated common gardens across gradients of (a) soil temperature and (c) soil moisture. Each line represents a reaction norm of an individual genotype in response to soil temperature or moisture in the common garden, colored by its source climate (principal component (PC) 1; blue = cooler & wetter, red = hotter & drier). (b) The sensitivity of genotype fitness to growing season soil temperature (the slope values from panel (a)) plotted against the genotypes' source climate. (d) The sensitivity of genotype fitness to growing season soil moisture (the slope values from panel (c)) plotted against the genotypes' source climate. In panels (b, d), each dot represents a genotype and is the linear sensitivity of a single, colored line depicted on the associated graph to its left calculated using the minimum and maximum of the environmental range. Dashed lines on the scatterplots represent lines of best fit with shaded bars representing 95% confidence intervals.

[−0.16, 1.10]). There was stronger evidence of genotype-by-temperature interactions for fecundity (Fig. S5c): while nearly all genotypes, on average, exhibited declines in fecundity with increasing soil temperature (NB regression coefficient $\gamma_{\text{temp}} = -0.19$ [−0.84, 0.47]), the magnitude of this decline varied widely (Fig. S5d). Genotype-by-temperature interactions for fecundity resulted in similar variation in fitness responses across genotypes (Fig. 3a), albeit in a different direction, on average, with most genotypes exhibiting increased fitness as soil temperature increased (Fig. 3b; most points above the horizontal line at 0).

Across the soil moisture gradient, we found stronger evidence of genotype-by-environment interactions for survival (Fig. S6a) and weaker evidence of genotype-by-environment interactions for fecundity (Fig. S6c). Consistent with responses to soil temperature, genotype-by-moisture interactions contributed to non-significant overall effects of soil moisture on survival and fecundity ($\beta_{\text{vwc}} = -0.39$ [−1.06, 0.27]; $\gamma_{\text{vwc}} = 0.27$ [−0.33, 0.85]). There was considerable variation in the magnitude and direction of responses (Fig. 3c), but most genotypes exhibited increases in fitness with increases in soil moisture (Fig. 3d; most points above the horizontal line at 0).

Fig. 4 Evidence consistent with local adaptation from a replicated common garden experiment with 96 *Bromus tectorum* genotypes. 'Climate mismatch' represents the difference between the weather a plant experienced during the experiment (calculated for each common garden site and year combination) and the source climate of that plant's genotype (30-yr norm). The differences in climate were calculated based on the first principal component of 19 bioclimatic variables and represented a continuum from hotter and drier to cooler and wetter climates. There are significant declines in (a) fitness (calculated as the probability of survival to reproduction $\times$ the number of seeds produced), (b) survival, and (c) fecundity as climate mismatch deviates from 0. Bolded lines and shaded envelopes represent predicted model means and 95% credible intervals, respectively. Dashed black vertical lines denote when climate mismatch = 0, and colored lines denote the value of climate mismatch when the demographic metric of interest is maximized and are colored by its position on the x-axis relative to 0. Points on (b, c) represent raw data.



Is genotype response to climate related to climate of origin?

In a *post hoc* analysis relating the environmental sensitivity of each genotype in the common garden to its source climate as characterized by PC 1, we found weak evidence to support local adaptation to soil temperature (Fig. 3b) and no evidence to support local adaptation to soil moisture (Fig. 3d). On average, genotypes from hotter and drier source locations experienced the greatest gains in fitness as soil temperature increased in the common garden, whereas genotypes from cooler and wetter source locations experienced smaller gains in fitness as temperature increased (sensitivity = -0.002 [-0.005 , 0.001]; Fig. 3b). This pattern was primarily driven by fecundity (sensitivity = -0.002 [-0.004 , 0.000]; Fig. S5d). While this effect was not statistically significant, it aligns with ecological intuition: genotypes from cooler and wetter environments experience the greatest declines in seed production in response to increasing temperatures. By contrast, there was no clear pattern of genotype sensitivity to soil moisture as a function of source environment across survival (sensitivity = -0.047 [-0.238 , 0.144]; Fig. S6b),

fecundity (sensitivity = -0.025 [-0.132 , 0.082]; Fig. S6d), and fitness (sensitivity = -0.057 [-0.236 , 0.122]; Fig. 3d).

We found stronger support for local adaptation to climate in the introduced range when we used the 'climate mismatch' metric to explain variation in plant demographic rates (Fig. 4). There was a strong negative quadratic relationship between climate mismatch and fitness that peaked at the environmental optimum of climate mismatch = 0 (derived quadratic curvature = -0.309 [-0.605 , -0.138]; Fig. 4a). We found this pattern for both survival ($\beta_{\text{climmis2}} = -0.095$ [-0.135 , -0.056]; Fig. 4b) and fecundity ($\gamma_{\text{climmis2}} = -0.061$ [-0.084 , -0.038]; Fig. 4c), suggesting both underlying demographic rates contribute to the relationship observed for the composite metric of fitness (Fig. 4a). Fitness peaked, on average, when climate mismatch was higher than zero. However, this mismatch value was not significantly different than zero (climate mismatch at maximum fitness = 0.629 [-0.458 , 1.718]), suggesting strong evidence for local adaptation to the source climate. This trend was driven by survival (climate mismatch at maximum survival = 1.600 [-0.221 , 4.128]; Fig. 4b), with hotter and drier conditions, on average, increasing survival. By contrast, fecundity was

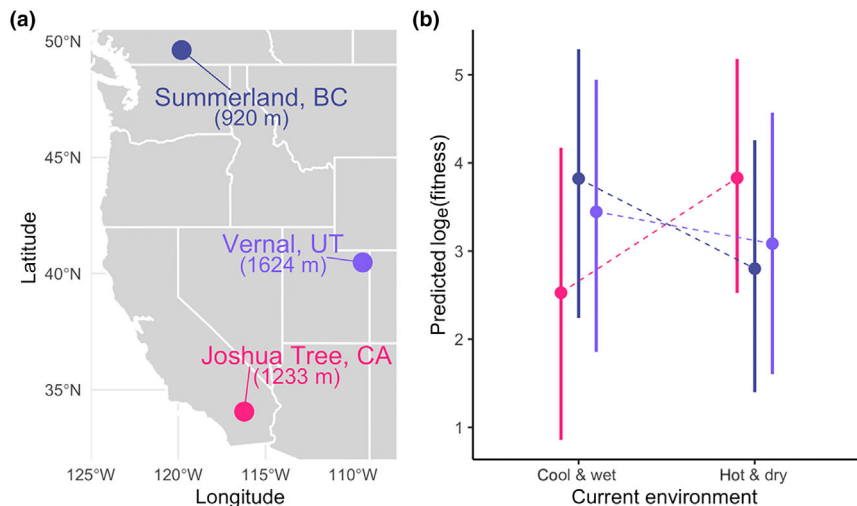


Fig. 5 Predicted log fitness (calculated as the probability of survival to reproduction $\times$ the number of seeds produced) for three *Bromus tectorum* genotypes in a cool and wet environment within the experiment (Cheyenne, WY, 2023, white gravel) and in a hot and dry environment within the experiment (Wildcat, ID, 2022, black gravel). The genotypes originated from field-collected populations in Joshua Tree, CA, Summerland, BC, and Vernal, UT, shown on the map in (a). Predictions consider weather and climate mismatch covariates and are averaged across density treatments. Points in (b) represent predicted means, and error bars represent 95% credible intervals.

maximized when the weather in the common garden was virtually identical to genotypes' source climates (climate mismatch at maximum fecundity = -0.120 [$-1.251, 1.031$]; Fig. 4c).

Together, the effects of climate mismatch, genotype-by-temperature, and genotype-by-moisture interactions resulted in meaningful differences in fitness across the environmental gradients we evaluated in this study. For example, in Fig. 5 we show the predicted fitness for three select genotypes that range across our PC 1 climate axis (hot and dry = Joshua Tree, CA; moderate = Vernal, UT; cool and wet = Summerland, BC) in both the coolest and wettest (white gravel, Cheyenne, WY, 2023) and the driest and hottest (black gravel, Wildcat, ID, 2022) observed soil microclimate conditions. We specifically chose these genotypes to highlight the potential magnitude of differences across our source location range. Both the Joshua Tree, CA and Summerland, BC genotypes exhibited a preference for environments that are like their source environment. In fact, our model predicts that the Joshua Tree genotype would produce, on average, 3.5 times more seeds in a hot, dry environment than in a cool, wet environment (assuming average density). The Summerland genotype would produce 2.8 times more seeds when in a cool and wet environment compared with a hot and dry environment.

Is there evidence that ecological or evolutionary responses to climate are mediated by intraspecific density dependence?

Density dependence mediated both ecological and evolutionary responses of *B. tectorum* to climate (Fig. 6). Averaged across soil temperature and moisture, plants at high densities exhibited higher survival and lower fecundity than those at low densities (Fig. 6a,b,d,e). Additionally, plant survival at high densities was less sensitive to soil temperature and more sensitive to soil moisture than was survival at low densities (Fig. 6a,b; $\beta_{\text{temp} \times \text{dens}} = 0.126$ [$0.011, 0.242$], $\beta_{\text{vwc} \times \text{dens}} = 0.126$ [$0.003, 0.246$]). The effect of soil temperature and moisture on fecundity exhibited

less density dependence (Fig. 6d,e). When survival and fecundity responses were aggregated to a composite fitness metric, there was no clear density-dependent response to climate (Fig. 6g,h).

Evidence to support local adaptation to climate for *B. tectorum* was also density dependent. For survival, the strength of support for local adaptation (i.e. the curvature of the parabola) was invariant to density (Fig. 6c; $\beta_{\text{clim} \times \text{dens}} = -0.011$ [$-0.039, 0.016$]). In the high-density treatment, survival was maximized in a current environment that was hotter and drier than the source environment (Fig. 6c). For fecundity, the strength of evidence supporting local adaptation depended more on density, specifically with the strength of evidence increasing (i.e. parabola narrowing) at lower density (Fig. 6f; $\gamma_{\text{clim} \times \text{dens}} = 0.010$ [$-0.005, 0.025$]). Together, this resulted in density-dependent fitness responses that strengthened the support for local adaptation at low densities and shifted the climate mismatch value at maximum fitness at high densities (Figs 6i, S7; difference in curvature = 0.003 [$0.000, 0.007$]; difference in climate mismatch at maximum fitness = 1.215 [$0.040, 2.575$]).

How does fitness change across source populations under elevated temperature?

Our model simulation of a 2°C increase in temperature across genotype source locations predicted that most *B. tectorum* genotypes (76 out of 96 total genotypes) would experience increases in fitness under increased temperature, but the magnitude of the predicted increase varied widely (Fig. 7). Some genotypes in source environments that were already very warm exhibited predicted decreases in fitness; however, these decreases were generally smaller in magnitude compared with genotypes predicted to experience fitness increases. The greatest increases in fitness were mostly predicted for genotypes in cooler and wetter source environments. These results align with general patterns across genotypes in Fig. 3a: when starting at a low source location temperature, fitness curves generally sloped upward and when starting at a high temperature, fitness curves generally sloped

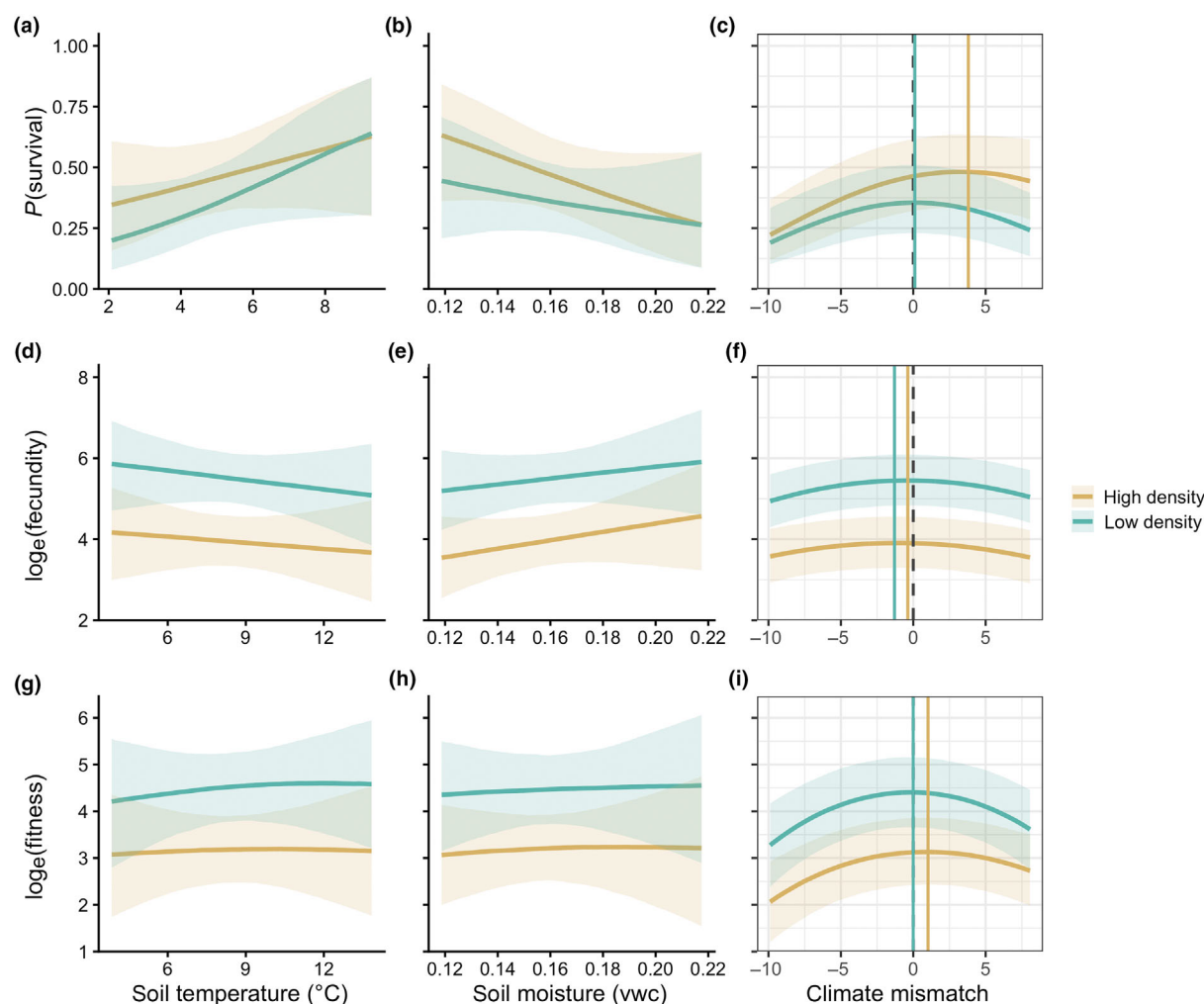


Fig. 6 *Bromus tectorum* survival (a–c), fecundity (d–f), and fitness (g–i; calculated as the probability of survival to reproduction $\times$ the number of seeds produced) as a function of density (colors) and soil temperature (a, d, g), soil moisture (b, e, h), or climate mismatch (c, f, i; the difference between the weather a plant experienced during the experiment and the source climate of that plant's genotype) in a replicated common garden experiment. Soil temperatures for survival (a) were average daily temperatures between January and May, and soil temperatures for fecundity and shown for fitness (d, g) were average daily temperatures between March and May. Lines represent predicted model means, and shaded bands represent 95% credible intervals. Grids in (c, f, i) were added to help visually compare differences in curvature.

downward. However, failing to account for eco-evolutionary processes would have led to incorrect predictions of the direction of fitness change given a 2°C increase in temperature for 15 out of 96 genotypes (Fig. S8).

Discussion

Despite decades of ecological and evolutionary research on the abiotic and biotic drivers of population dynamics, predicting plant fitness responses to climate change remains a major challenge. Using demographic data from 96 *B. tectorum* genotypes grown at four common garden sites across two growing seasons, we show that improved prediction may require accounting for complex interactions between climate, local adaptation, density dependence, and fitness. Context dependence is often a hallmark

of ecology (Catford *et al.*, 2022), and accounting for it may be essential for predicting plant fitness under climate change.

Fitness represents opposing responses of survival and fecundity

We found that survival and fecundity responded in opposing ways to environmental variation, meaning that using only one of them as a proxy for fitness would be misleading. A meta-analysis by Anderson (2016) similarly found that failing to account for binary fitness outcomes (e.g. survival) could bias estimates of plant fitness in response to climate change. Interestingly, we found stronger evidence of genotype-by-environment interactions for fecundity than survival, suggesting that heritable responses to climate change may differ for these two component

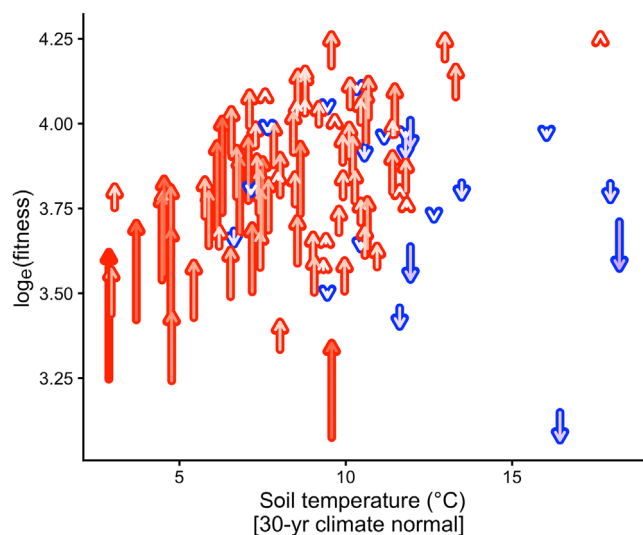


Fig. 7 Predicted change in log fitness (calculated as the natural-log of the probability of survival to reproduction $\times$ the number of seeds produced) for *Bromus tectorum* genotype source populations given a 2°C increase in temperature. Each arrow represents a genotype from the common garden experiment at their source location (assuming that genotypes are locally adapted to their source climates) and are sorted by their average temperature on the x-axis. The length of the arrow represents the model predicted increase (red) or decrease (blue) in fitness given a 2°C increase in temperature. The hue of the arrow also represents the magnitude of the change (i.e. darker arrows = larger changes). Model predictions were made using average soil moisture and density across the common garden sites, years, and treatments.

vital rates of fitness. Foundational work comparing trait heritabilities to ‘true’ measures of fitness further highlights this phenomenon (Merilä & Sheldon, 2000). Future studies should carefully consider which traits to use as proxies for performance as this choice could influence predictions of evolutionary responses to climate change.

Although seemingly counterintuitive, the decreases in survival with increasing soil moisture in our common gardens may reflect the effects of winter snowpack more than spring precipitation, as *B. tectorum* seeds were planted in fall and overwintered *in situ*. Thus, as *B. tectorum* is a winter annual, some seeds may have emerged in the fall and then died under snowpack before our spring census. Griffith & Loik (2010) and Compagnoni & Adler (2014) similarly found that *B. tectorum* population growth rates decreased with increased snowpack due to declines in seedling emergence and survival. Regardless of the mechanism, after combining probability of survival with fecundity to estimate fitness, we found no overall monotonic relationship between the measured abiotic variables and fitness.

Evidence for local adaptation to climate

Bromus tectorum genotypes varied in their response to gradients of soil temperature and moisture despite, or perhaps due to, a lack of overall fitness response to changes in their microclimate. Evidence that these genotype-by-environment interactions were

related to genotypes’ source climate provided strong support for local adaptation to climate in the introduced range. Our study builds on previous work that provided evidence for local adaptation in *B. tectorum* introduced populations (Leger *et al.*, 2009; Hufft & Zelikova, 2016; Gamba *et al.*, 2025; Vahsen *et al.*, 2025). Variation in genotypic sensitivities to soil temperature and moisture found in this study may provide insight into the most important selective forces in the introduced range. Temperature may be a more predictable selective agent than moisture, because climate models predict strong increases in temperature with more confidence than changes in soil moisture (IPCC, 2023). Thus, adaptation to temperature may be a more likely evolutionary mechanism by which introduced populations persist. This stronger signal of temperature-induced local adaptation may also reflect evolution of plasticity, as previous work has shown that plants from warmer, lower-elevation environments tend to be more plastic than those from colder, high-elevation climates (e.g. Emery *et al.*, 1994; Vitasse *et al.*, 2013; Schmid *et al.*, 2022; but see Münzbergová *et al.*, 2017).

Fitness systematically declined with an increasing difference between the weather *B. tectorum* genotypes experienced in the common garden and their source climates. This finding is consistent with a recent genomic analysis of introduced and native populations of *B. tectorum* (Gamba *et al.*, 2025). However, because we did not use a reciprocal transplant design, there could indeed be additional agents of selection that are unrelated to climate but have similar spatial structure across source populations. We did compare climate effects across common garden environments and surface temperature (gravel) treatments, which partly mitigates this concern (Anderson, 2016), but recent attempts to make predictions with a similar approach to ours raised concerns because selection regimes can shift from year to year (Villoutreix *et al.*, 2025). More work is needed to predict the effects of increasingly variable environmental conditions on fitness.

Density as a mediator of environmental and evolutionary drivers

Density dependence had a strong effect on fitness overall but was not an important mediator of climate change effects, contrary to our expectations. The opposing effects of the component vital rates on fitness suggest a trade-off between maximizing survival and fecundity to maintain consistent fitness under a variety of environmental conditions. These opposing effects perhaps also weakened the density dependence of sensitivities to climate. It is possible that the high plasticity observed in *B. tectorum* could be at play here (Hufft & Zelikova, 2016; Vahsen *et al.*, 2025), such that plants may be able to adjust their life-history strategy based on their perceived biotic and abiotic environment to reduce fitness variability across years. Additionally, plants at high density can alter the microenvironment around them (Gates, 1965; Antonovics & Levin, 1980), which could mediate the relative effect of soil temperature treatments. Our recent finding that *B. tectorum* flowers earlier at high densities compared with low densities aligns with this hypothesis (Vahsen *et al.*, 2025). Interestingly, we did find evidence supporting that local adaptation to

climate was stronger for plants at low density compared with plants at high density. This pattern may have arisen because, at low density, the reduced effect of competition allowed for more distinct genetic differences to arise (Weiner, 1990). At high density, individuals may alternatively need to put more resources into competing. In effect, competition may 'swamp' the influence of local adaptation (Siepielski *et al.*, 2016). Additionally, high-density conditions may favor plasticity. If plants shift their life-history strategy in response to increased intraspecific density, the relative benefit of being locally adapted would likely decrease.

Interactions between local adaptation and density dependence could be particularly important for invasive species. As invasive populations expand their range, they generally shift from high population densities in their core to low densities at the expanding edge. If local adaptation is the strongest at low densities, as in this experiment, then fitness may show steeper declines as plants encounter more novel habitats when they arrive at lower densities. Disentangling the genetic and demographic drivers of population dynamics on range edges is, however, still a challenge (Hargreaves & Eckert, 2019). Indeed, trade-offs between dispersal and resource acquisition traits can even result in increases in densities at range edges (Fronhofer & Altermatt, 2015). The potential for local adaptation to occur at range edges also depends on the standing genetic diversity, which can decrease because of spatial sorting or bottleneck effects (Eckert *et al.*, 2008; Pierce *et al.*, 2017). Finally, we discovered that the observed increase in fitness under hotter and drier conditions relative to seed source locations only occurred at high densities, further suggesting that density regulates how populations respond to climate. We speculate that *B. tectorum* fitness may increase in existing populations where density is high if climates become hotter and drier, due to increased overwinter survival.

Predicting fitness under climate change

Our predictions align with previous empirical work which found that increased temperature has a consistently positive effect on *B. tectorum* fitness and population growth rate (Compagnoni & Adler, 2014; Blumenthal *et al.*, 2016). The extensive sampling of genotypes across Western North America coupled with exposure to a continuous gradient of different temperatures in our multi-year, multi-site common garden experiment helps confirm the generality of these previous findings while introducing important context dependence. Our results show the importance of eco-evolutionary context in determining the direction and magnitude of fitness responses to climate change, although future work may lead to more realistic projections of cheatgrass population growth under specific climate change scenarios. Importantly, in addition to soil temperature and moisture, fire is another abiotic driver that has well-recognized feedbacks with *B. tectorum* demography (Pellant, 1996). It will be important to investigate how 'pulse' disturbances interact with climate change, density, and genetic variation to shape demographic outcomes.

Here, we have illustrated the challenge of predicting plant fitness under climate change, particularly in invasive species like *B. tectorum* that can quickly dominate ecosystems in which they

invade. To predict plant fitness, knowing how the abiotic environment could change is not enough; we also need to know how genetic variation and density could change. Put plainly, there was no simple relationship between *B. tectorum* fitness and either soil temperature or moisture. Rather, these effects were genotype- and context-specific. Characterizing genotype-by-environment interactions and density-dependent responses to evolutionary context was necessary to understand how *B. tectorum* fitness responds to climate. An important next step will be to explore how generalizable these patterns are across species and communities as interspecific interactions could explain additional variation in fitness responses to climate change (Kooyers *et al.*, 2025).

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Competing interests

None declared.

Author contributions

MLV, DMB, CB, MJG, MBH, JRL, EAL, LMP and PBA planned and designed the research. MLV, JJV, DG, TMM, NP, SJR, DMB, EAL, LMP and PBA conducted the experiments and analyzed the data. TMM, RAH, RH, DMB, CB, MJG, MBH, JRL, EAL, LMP and PBA acquired the funding to support the work. MLV and PBA wrote the original manuscript draft. All authors edited the manuscript.

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Data availability

All data and code used for the described analyses are available via a Zenodo archived Github repository doi: [10.5281/zenodo.20041620](https://doi.org/10.5281/zenodo.20041620).

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Source locations of genotypes.

Fig. S2 Relationship between inflorescence mass and seed count.

Fig. S3 Posterior predictive checks for hurdle model.

Fig. S4 Hurdle model accuracy and goodness-of-fit.

Fig. S5 Genotype sensitivities to soil temperature.

Fig. S6 Genotype sensitivities to soil moisture.

Fig. S7 Variable strength of local adaptation by density.

Fig. S8 Predicted change in fitness without eco-evolutionary context.

Methods S1 Calculating a continuous density metric.

Methods S2 Predicting change in fitness under future climate conditions.

Notes S1 Hierarchical hurdle model for fitness.

Table S1 Location and climate summaries for genotype source locations.

Table S2 Summary statistics for hierarchical hurdle model parameters.

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